



Introduction to Coastlines



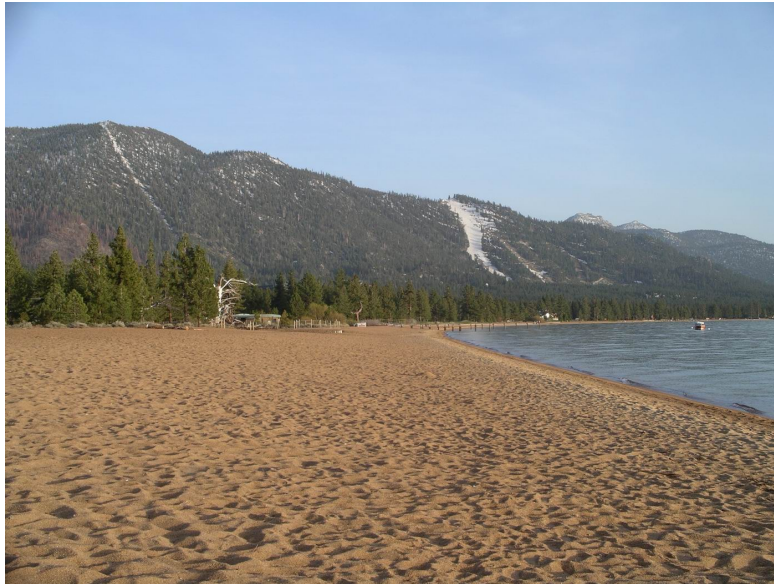
Coastlines

- Are a dynamic interface between the sea, the land and the atmosphere.
- Constantly being modified by wave activity.
- Experiences intense human activity.



Beach Sand

- Where does the sand originate from?
 - Streams
- Composition of sand?
 - Reflects the rock types the streams are eroding.
 - Typically quartz and feldspar along the California coastline from the erosion of the Sierra Nevada mountains.



Lake Tahoe Shoreline



Santa Cruz Shoreline

Other places....

- Florida Coastline
 - Northern and Central Georgia via Flint and Chattahoochee Rivers
 - Mostly pure clear Quartz
 - Clear Quartz having fewer impurities appears brighter
- Hawaii
 - Black sand beaches from Basalt
 - White beaches from the breakdown of coral
 - Green beaches from the mineral olivine



Panama City Beach, Florida

Black Sand Beach From 2003 Kilauea Lava Flow

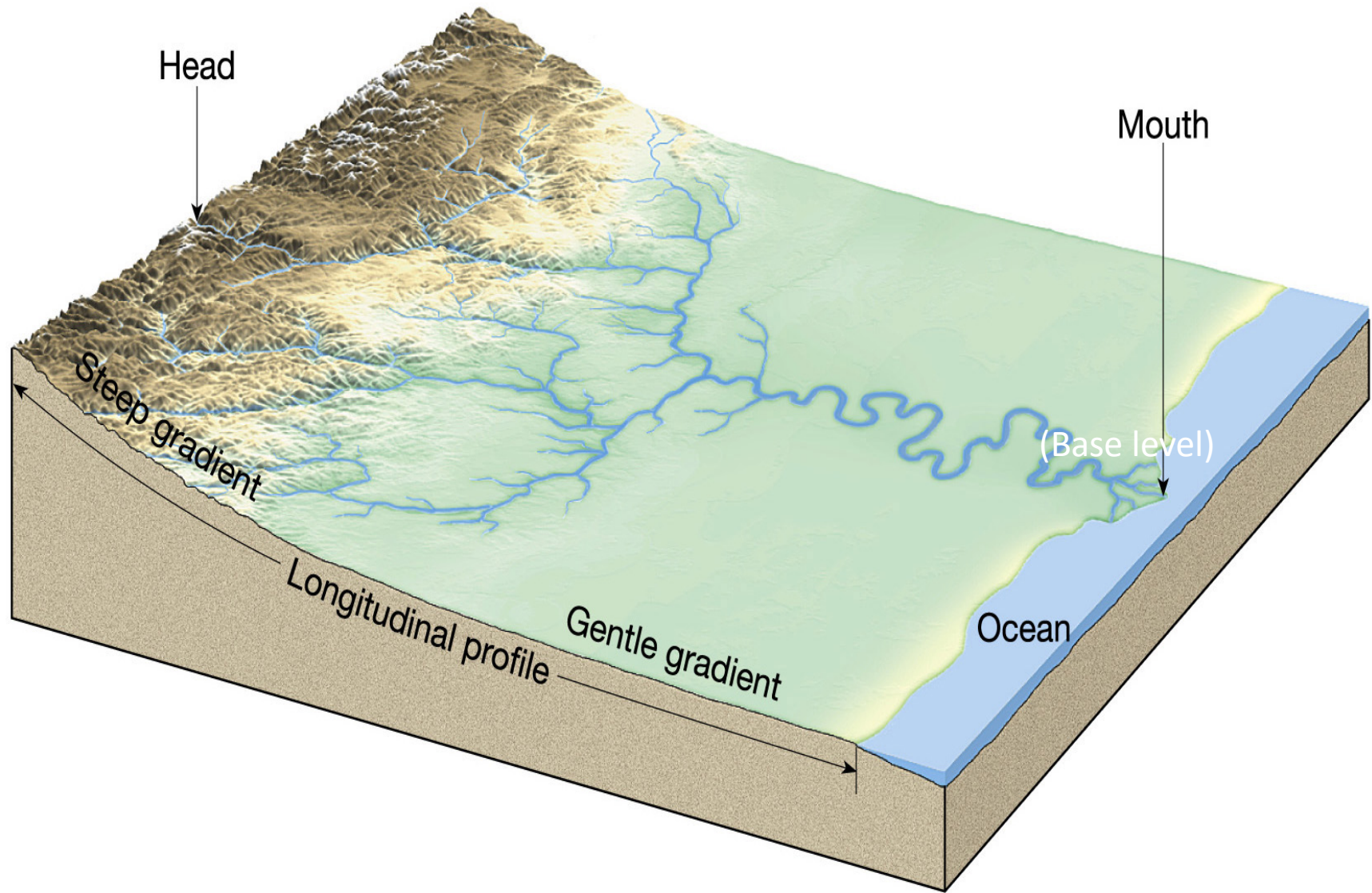




Marin Headlands



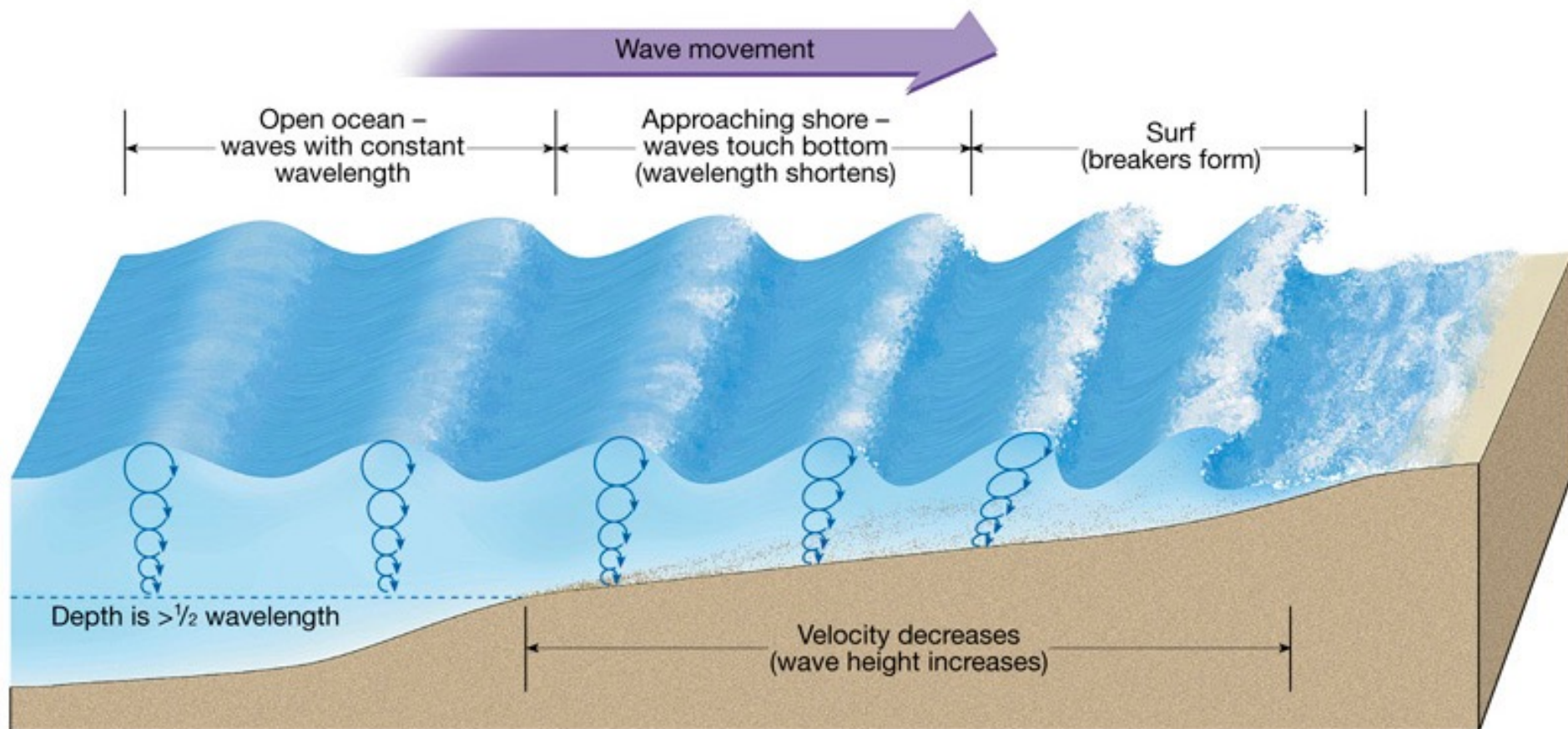
Longitudinal Profile of a Stream





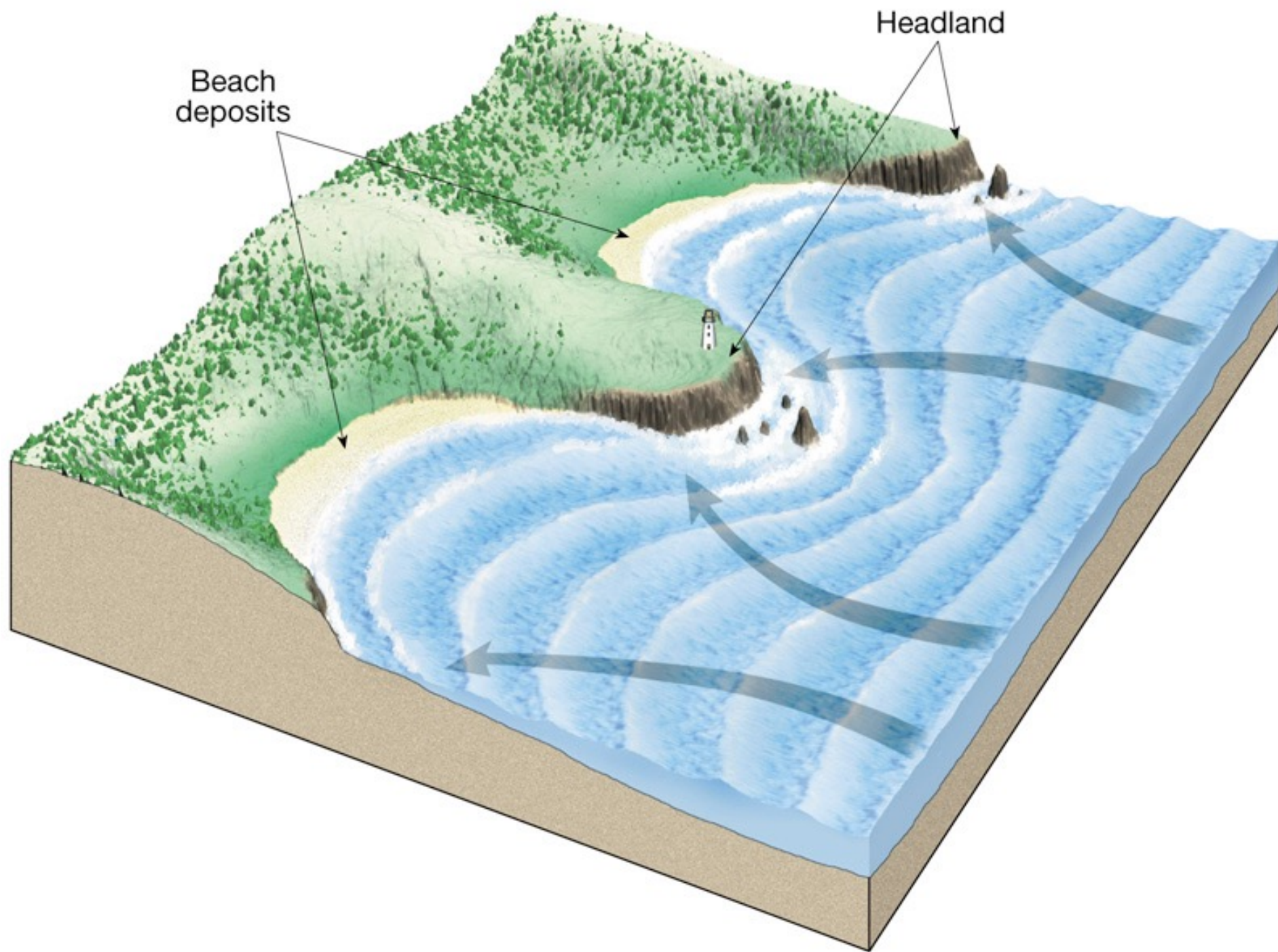
Coastal Processes





Wave Refraction: Bending of a Wave.







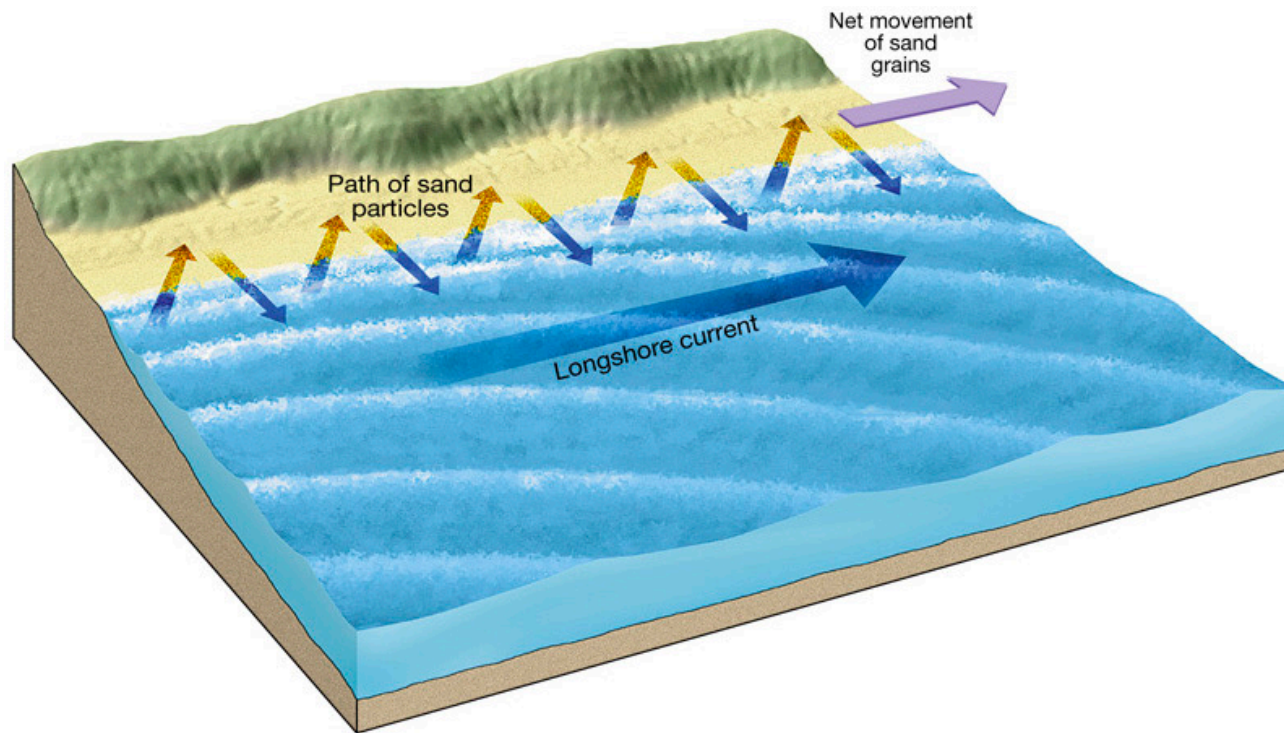
Longshore Current

- A current of water that runs parallel to the shoreline.
- Created by oblique waves
- Located in the “surf zone” where waves break.



Longshore Transport

- Sand that is transported parallel to the coastline in the longshore current.





Coastal Features



Shoreline Features

- Created by either EROSION or DEPOSITION
- These features vary depending on
 - The rock types of the shore
 - Sediment input from streams
 - Wave Intensity
 - Currents
 - Whether the coast is
 - Stable
 - Sinking
 - Rising

Features caused by wave EROSION

- Wave cut cliffs
 - Form when a rocky shore erodes at approx. the same rate.
 - Cliff moves landward as waves attack its base.



Wave cut cliff

Features caused by wave EROSION

- Sea caves
 - When waves attack weak portions of a cliff.



Sea cave

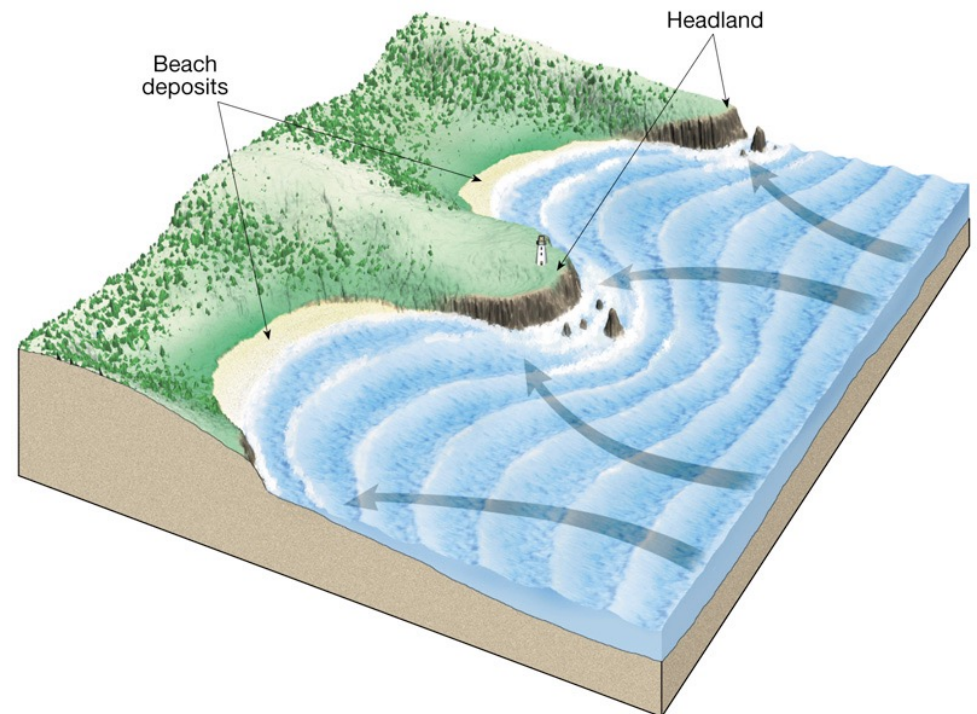
Features caused by wave EROSION

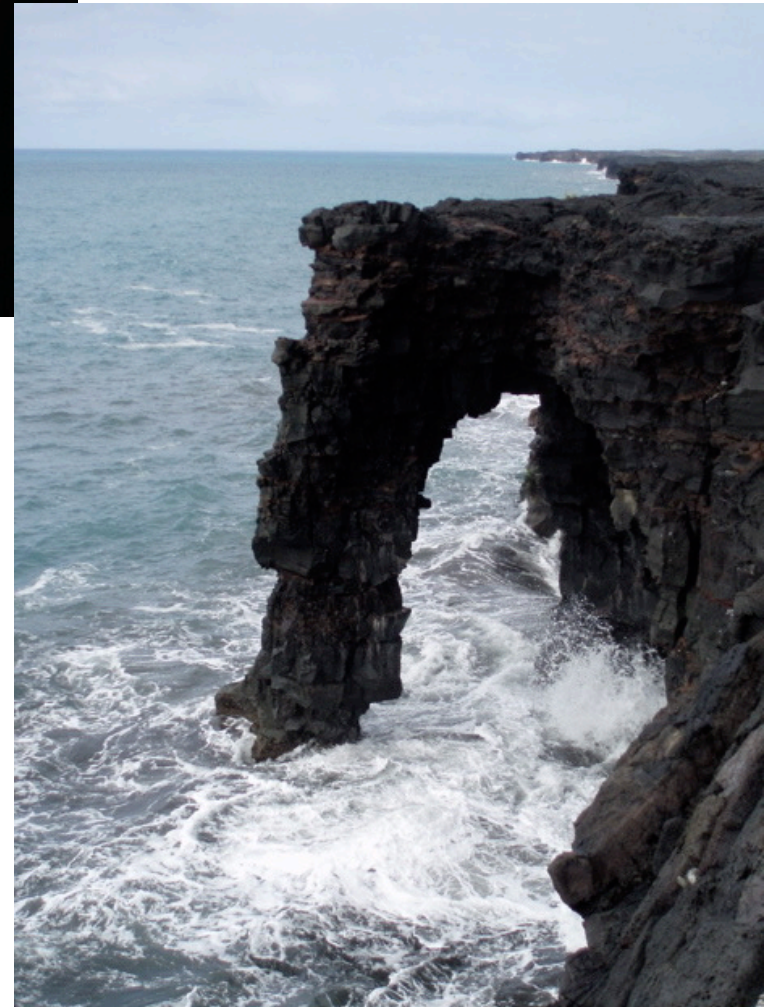
- Wave cut platforms
 - As erosion attacks a sea cliff, a platform is left behind.



Features caused by wave EROSION

- Sea arches
 - When waves completely cut through a sea cave resulting from headland erosion.
- Sea stack
 - Once a sea arch
 - Arch has collapsed

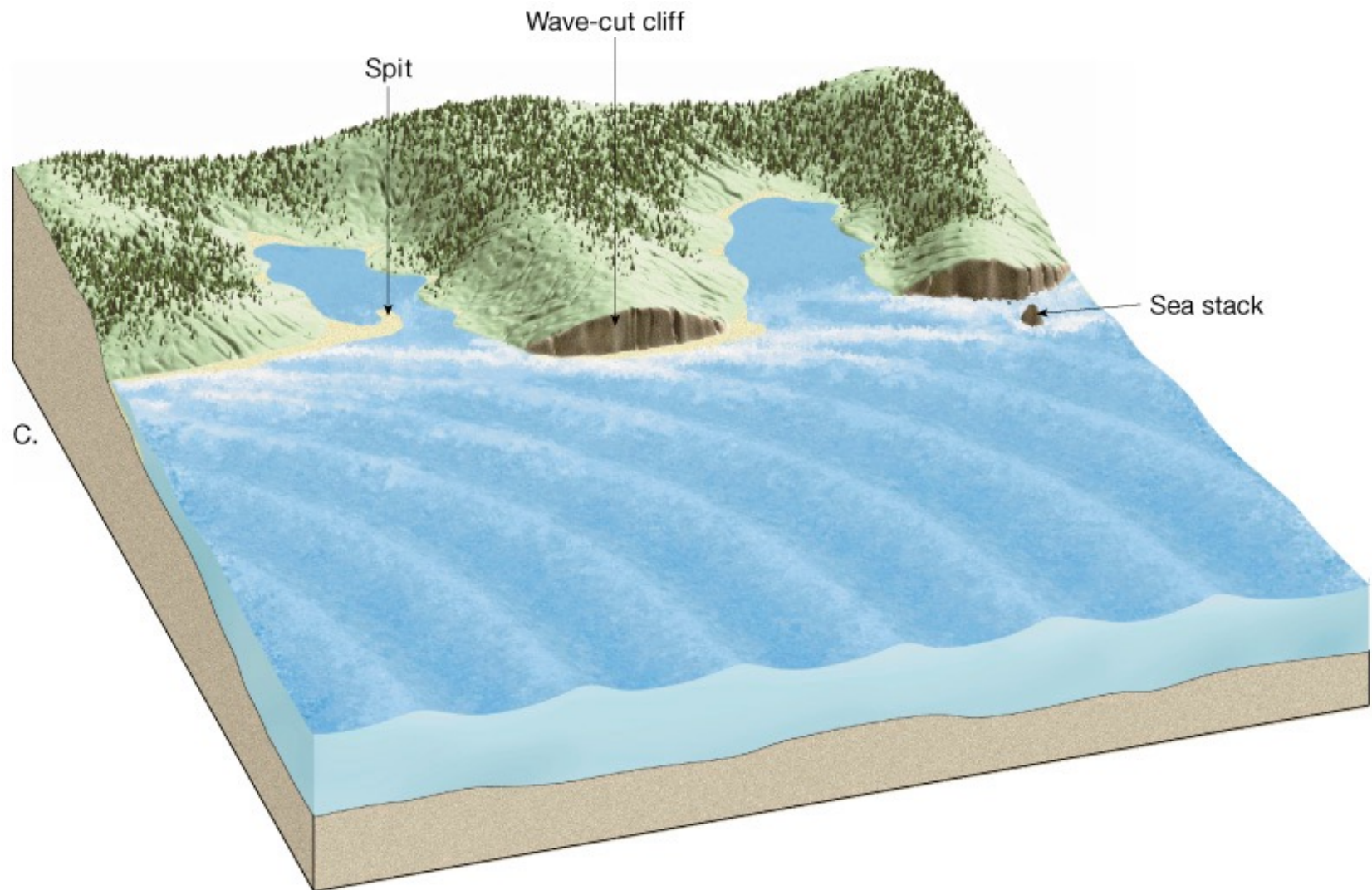




Features caused by wave DEPOSITION

- Spits
 - Elongated ridges of sand extending from land into the mouth of an adjacent bay.
 - Often the end of a spit hooks landward in response to wave generated currents
- Baymouth Bar
 - A sand bar that completely crosses a bay

Features caused by wave DEPOSITION



Features caused by wave DEPOSITION



Features caused by wave DEPOSITION



Features caused by wave DEPOSITION

- Tombolo
 - A ridge of sand that connects an island to a mainland or another island.





Coastal Hazards



Coastal Hazards

- Primary West Coast Hazards
 - El Nino & Frontal Storm Systems
 - Coastal Erosion and Subsidence
 - Earthquakes & Tsunami
 - Sea-level rise
- Primary East Coast and Gulf Hazards
 - Hurricanes
 - Coastal Erosion and Subsidence
 - Sea-level rise

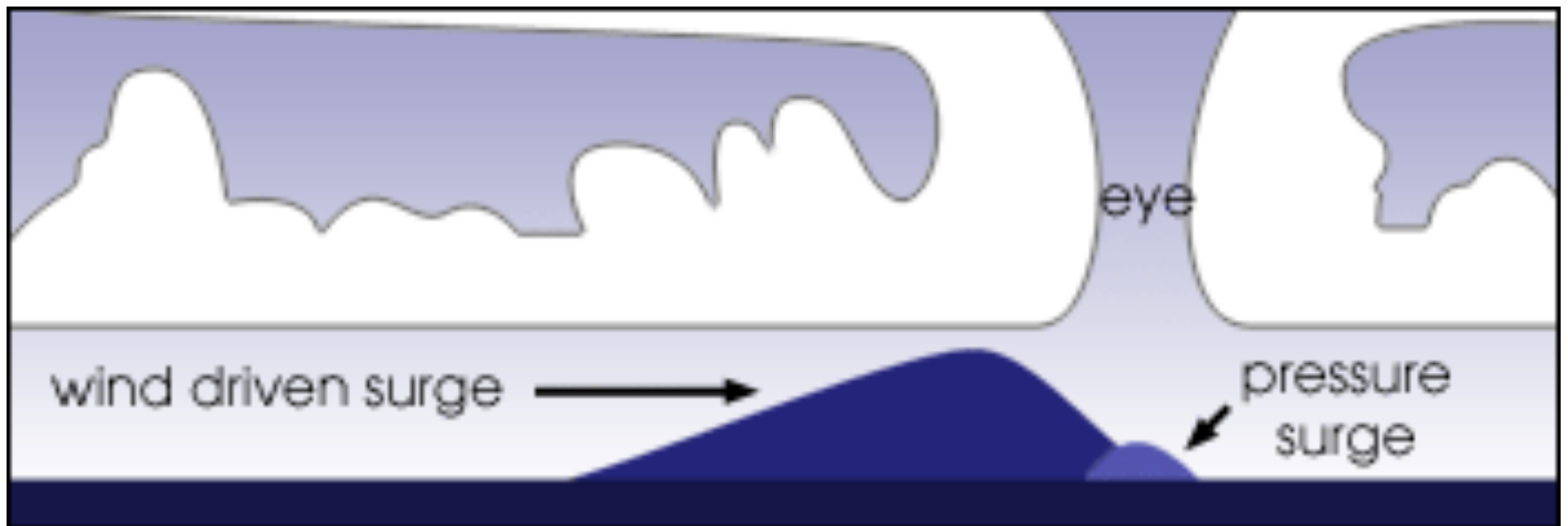
Storm Surge

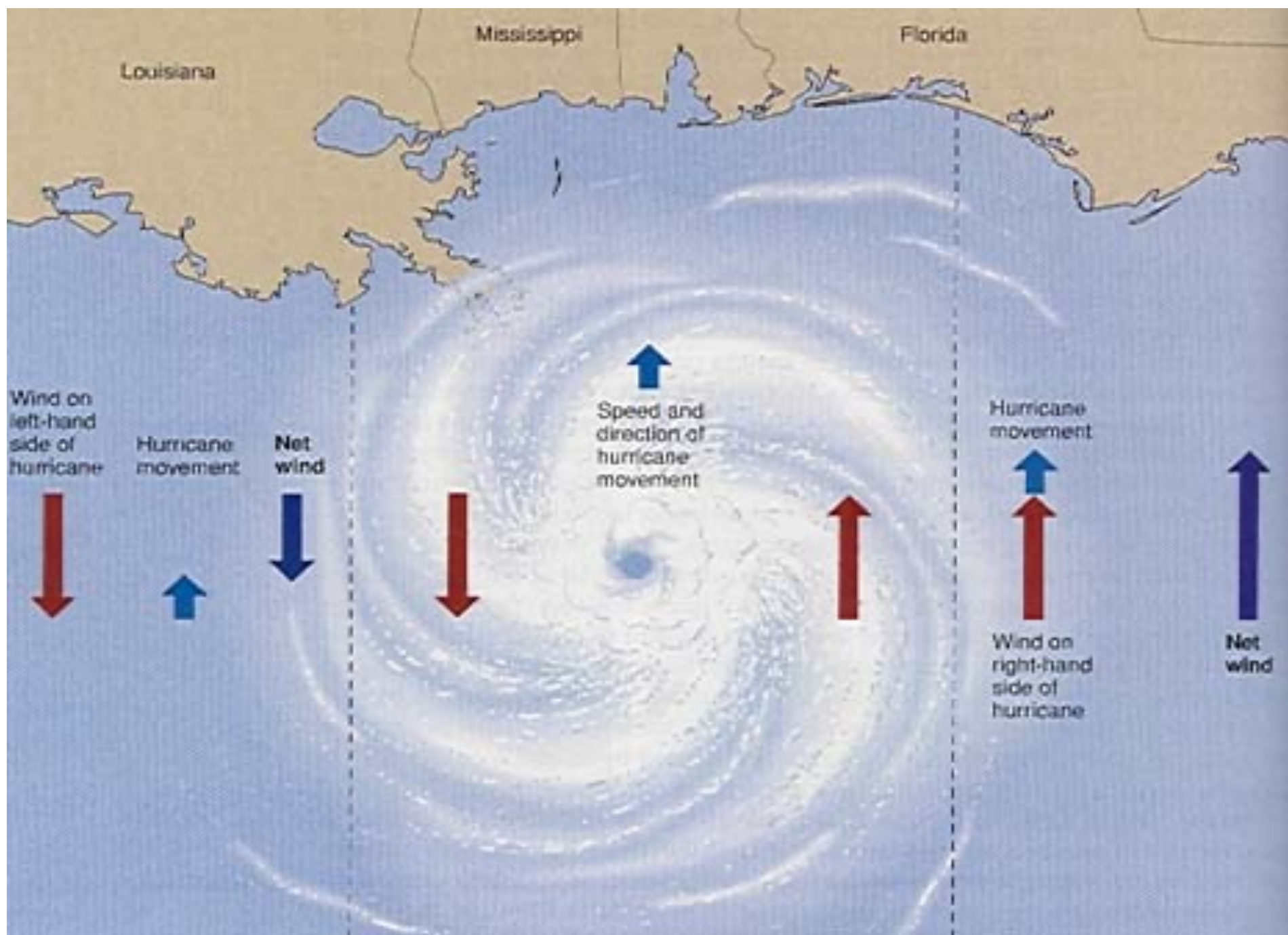
- A bulge of water that is driven to shore by either hurricanes or frontal storm systems.
- Dependant on:
 - Speed of the storm
 - Strength of the storm
 - High tide



What creates the “bulge”?

- A combination of two factors
 - Low central atmospheric pressure (small factor)
 - Surface convergence of air piling up water





Other factors that contribute to Storm Surge

- Speed of the storm
- Strength of the storm
- High tide



Katrina in Louisiana (2005)



A.



B.

Coastlines and sea-level change

- Eustatic Sea Level Change
 - Sea level change happens globally
 - Mostly due to water being added or removed from ocean basin.
 - Can also be due to underwater tectonic activity.
- Local Sea Level Change
 - Sea level change happens locally
 - Usually due to tectonic activity at a particular location along a coast.

Emergent Coasts

- Develop due to
 - Uplift of land (Local sea level change)
 - Coastal Faulting
 - Drop in sea-level (Eustatic sea level change)
 - Ice age
 - Water locked up in glacial ice
- Features seen
 - Uplifted wave cut cliffs and platforms

Emergent Coasts



Big Sur Coastline

Photo courtesy of Alan Glazner
Univ. of North Carolina, Chapel Hill

Emergent Coasts



Photo courtesy of Alan Glazner
Univ. of North Carolina, Chapel Hill

Emergent Coasts



Photo courtesy of Alan Glazner
Univ. of North Carolina, Chapel Hill

Submergent Coasts

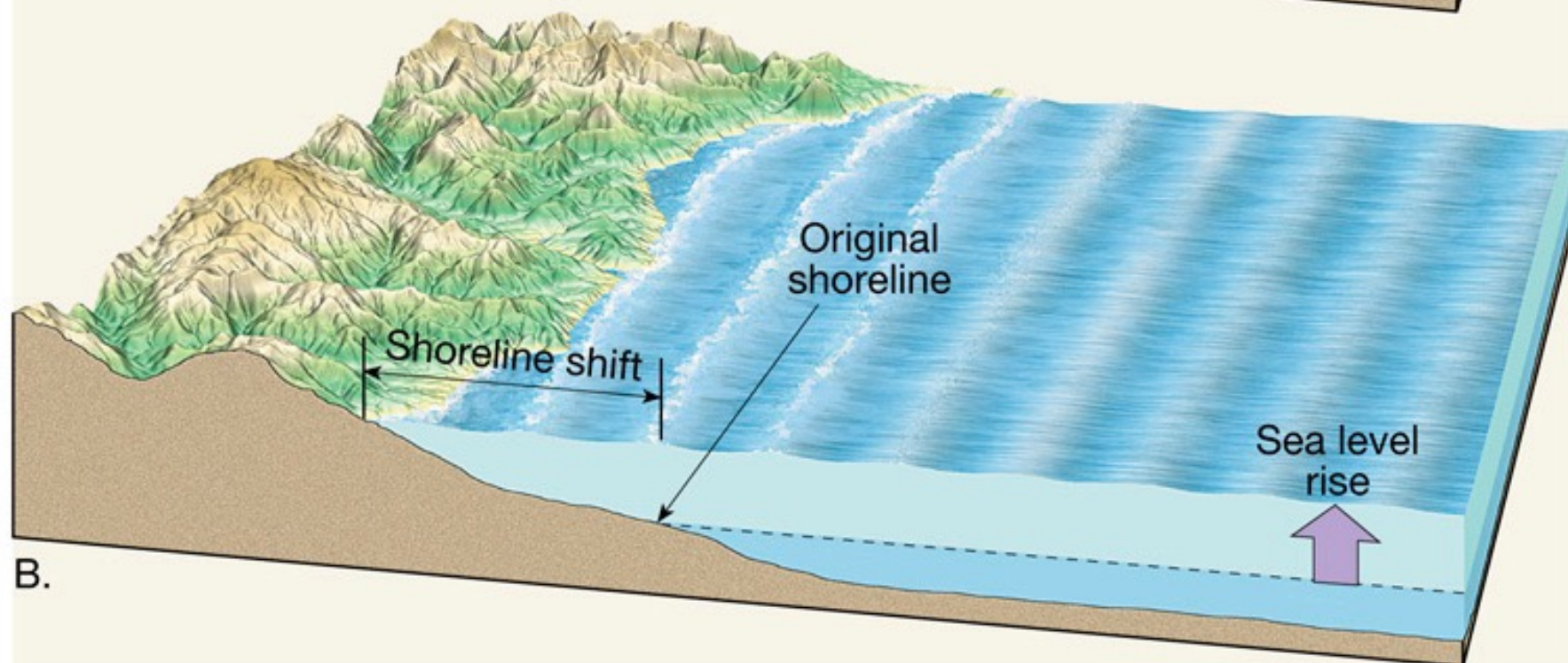
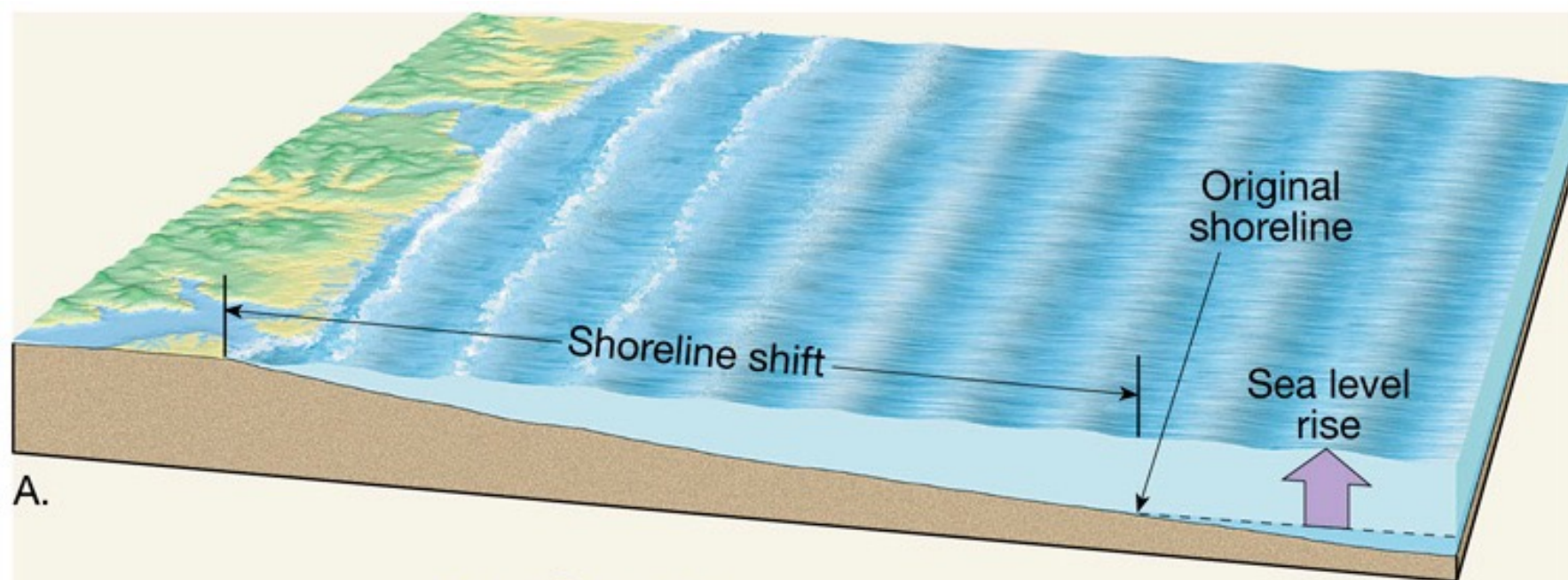
- Submergent Coasts
 - Develop due to
 - Subsidence
 - Weight
 - Faulting
 - Rise in sea-level
 - Melting of glaciers



Submergent Coasts

– Features seen

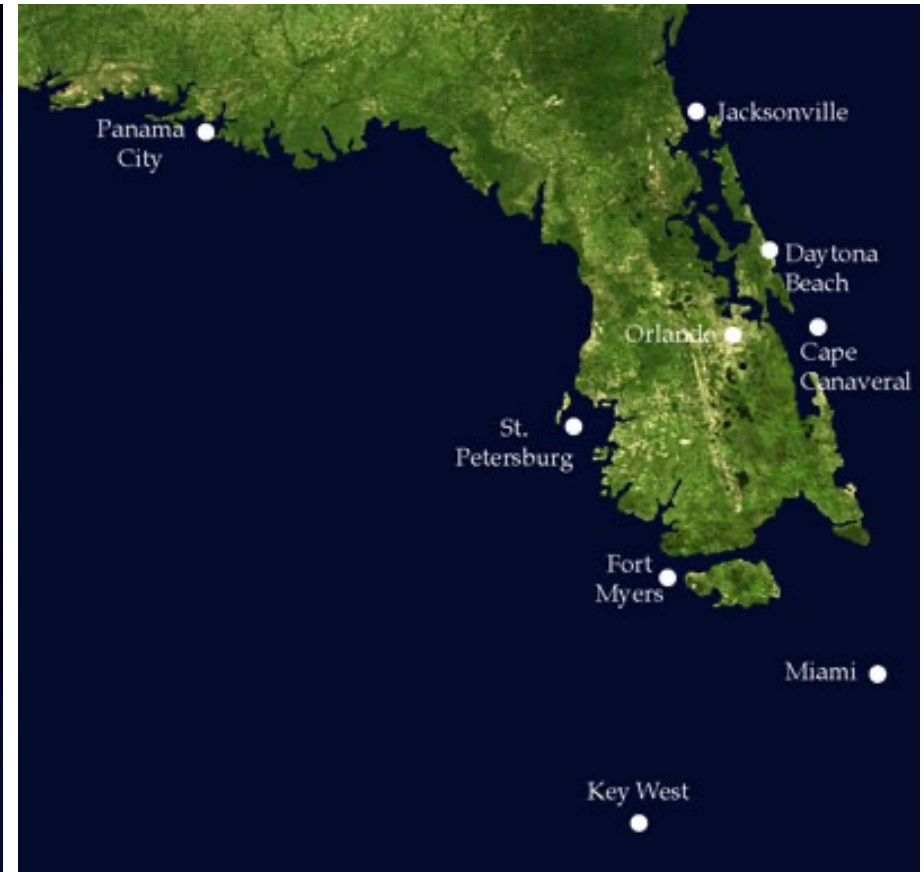
- Highly irregular shoreline
- Estuaries
 - A body of water partially surrounded by land where sea water mixes with fresh water.
- Drowned River Mouth
 - An estuary created by sea-level rise as sea water drowns a river mouth.



Today



20 Feet of Sea-Level Rise



Today

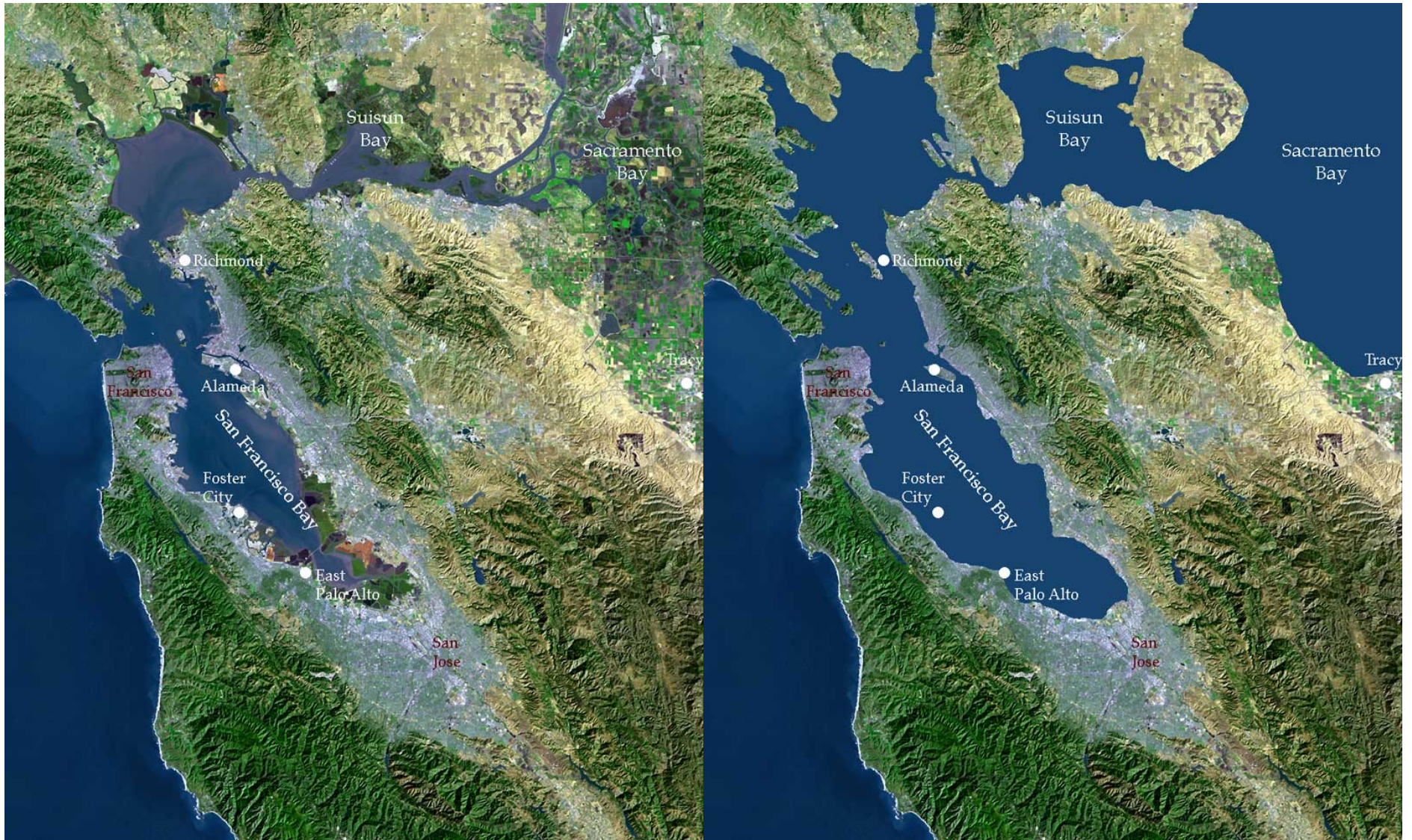


20 Feet of Sea-Level Rise



Today

20 Feet of Sea-Level Rise



Devils Slide



History of Devil's Slide

- 1890 – First road built through area.
- In 1920 – Railroad track put in place.
- In 1936 – Hwy 1 was built in former location of railroad track.
- 2005 – Tunnel construction begins to bypass slide. Completed in 2013.
- 2013 - Old hwy 1 will start to be used as a bicycle/hiking trail.